

Design and Evaluation of a Context-Aware Multimodal Recommendation and QA System with Retrieval-Augmented Generation

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Abstract

This paper proposes a novel design for a multimodal, context-sensitive recommendation and question-answering (QA) system, MMCARQA that integrates Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs). The architecture enables dynamic understanding and reasoning over diverse input modalities—including text, images, audio, video, and documents—thereby enhancing the relevance, personalization, and contextual accuracy of system responses. Drawing inspiration from security-aware architectures in IoT and federated AI systems, the framework emphasizes modularity, privacy, and scalability, supporting deployment across both edge and cloud environments. A detailed architectural diagram is presented, and key components such as multimodal preprocessing pipelines, adaptive retrieval strategies, memory-augmented generation modules, graph-based reasoning agents, and LLM-driven response generation are discussed in depth. To rigorously validate system performance, a comprehensive experimental evaluation methodology is proposed. The framework is benchmarked across multiple dimensions, including latency, retrieval-free accuracy, multi-hop graph reasoning, explainability, differential privacy guarantees, and energy efficiency on heterogeneous hardware platforms. Evaluation targets include sub-20 ms on-device classification, ≥ 0.85 precision@5 on QA tasks, $F1 \geq 0.75$ on structured reasoning datasets, and strong user trust ratings in interpretability studies. Together, the proposed design and its empirical evaluation demonstrate a scalable, explainable, and privacy-preserving multimodal QA solution capable of adapting to real-world domain-specific scenarios.

I INTRODUCTION

Traditional QA and recommendation engines are predominantly text-centric and often fail to account for rich contextual signals available in real-world scenarios. Recent advances in Retrieval-Augmented Generation (RAG) and multimodal models have demonstrated improved factuality and reasoning but still rely mainly on textual inputs. Simultaneously, application domains such as healthcare, education and smart spaces demand systems capable of ingesting and reasoning over heterogeneous data sources—audio, video, sensor data, and images—to provide accurate and context-sensitive guidance. This paper introduces a comprehensive architecture that unifies multimodal preprocessing, context capture, RAG, and LLM-driven generation. Inspired by security-aware IoT authentication frameworks, the proposed design also incorporates privacy-preserving mechanisms and modular deployment across edge and cloud resources. To support multimodal and context-sensitive recommendation, the architecture leverages deep representation learning techniques such as CLIP, Whisper, and BLIP to extract semantically rich embeddings from audio, images, and video streams. These embeddings are fused with structured and unstructured text data using cross-attention mechanisms to capture nuanced contextual signals. A context-awareness layer dynamically prioritizes modalities based on environmental cues and user intent, enabling more adaptive responses. Furthermore, integration with RAG allows the system to retrieve knowledge grounded in real-world multimodal evidence, enhancing reasoning and factual accuracy across diverse application domains.

In clinical diagnostics, timely and accurate interpretation of diverse patient data is critical. The proposed multimodal system enables the integration of radiology images (e.g., X-rays, MRIs), symptom transcripts from doctor–patient interactions, and structured Electronic Health Records (EHRs) to generate diagnostic suggestions with high contextual fidelity. Vision encoders such as ViT or BLIP are used to extract salient features from medical imagery, while ASR models transcribe spoken clinical conversations. This multimodal input is fused using attention-based mechanisms to produce a unified patient representation. Large language models (LLMs) are then guided by retrieval-augmented generation (RAG) pipelines to synthesize evidence-based diagnostic hypotheses, ensuring that suggestions are both clinically relevant and explainable. Moreover, privacy-preserving computation through federated learning ensures compliance with data governance regulations such as HIPAA. In personalized learning environments, understanding both content and learner behavior is key to effective recommendation. This framework processes inputs from lecture videos, handwritten or typed class notes, and learner attention signals captured through eye-tracking or interaction logs. Video frames are interpreted using BLIP or Donut models, while textual content is parsed using OCR and document parsers. Learner context—such as engagement duration, topic interest, and performance history—is encoded as behavioral metadata. These diverse inputs are fused into a multimodal learner profile that informs the recommendation of targeted study materials or revision strategies. The system dynamically adjusts to changing learner needs and ensures recommendations are aligned with cognitive load and learning objectives.

Legal professionals often deal with complex, multimodal content such as recorded client conversations, scanned contracts, and supporting documentation. The proposed architecture captures audio discussions using ASR models like Whisper, while legal documents are parsed and semantically embedded using OCR in combination with transformer-based language encoders. A fusion layer integrates these modalities into a cohesive case context, enabling LLMs to summarize multimodal contract discussions into actionable legal insights or to-do lists. The attribution engine further enhances explainability by tracing outputs to specific clauses, speech fragments, or legal precedents, providing transparency crucial for compliance and litigation support in legal workflows. Smart home assistants must operate within dynamic environments, responding accurately to user commands while interpreting real-time contextual cues. This system fuses user voice commands, environmental sensor data (e.g., temperature, motion, lighting), and recent user behavior to deliver context-aware automation guidance. Speech inputs are transcribed and semantically interpreted, while sensor data is encoded into temporal-spatial features.

II RELATED WORKS

Recent advancements in automatic speech recognition (ASR) have positioned models like Whisper and Wav2Vec at the forefront of modern audio processing. Machacek et al. (2023) demonstrate how Whisper can be adapted into a real-time transcription system, highlighting its robustness across diverse accents, varying speech patterns, and noisy acoustic environments. Such capabilities make Whisper a strong candidate for deployment in real-world, multilingual, and mobile scenarios. Similarly, Bain et al. (2023)

present WhisperX, which enhances time-alignment accuracy in long-form speech, a critical factor for applications in education, accessibility, and assistive technologies. These models deliver high transcription fidelity and are especially well-suited for integration into multimodal platforms that rely on accurate and real-time speech-to-text conversion as a foundation for downstream tasks like question answering or content summarization. The LangChain and Haystack frameworks have emerged as powerful orchestration tools for building Retrieval-Augmented Generation (RAG) pipelines, where contextually relevant information is retrieved and passed to large language models for response generation.

Chen et al. (2023) benchmark RAG pipelines with LLMs and offer detailed comparisons across different retrieval strategies, providing critical insights into performance tuning, scalability, and domain adaptation. Meanwhile, Burgan et al. (2024) detail the successful implementation of a RAG-based chatbot using LangChain, validating its extensibility and adaptability in educational applications such as tutoring, automated grading, and student support. LangChain's seamless integration with vector databases like Pinecone has proven instrumental in enabling scalable semantic search and advanced context management in RAG workflows. Matsuda and Frank (2024) describe how LangChain-based systems can extend beyond ChatGPT's default capabilities by supporting domain-specific vector retrieval, essential for precision in knowledge-intensive tasks like academic search or policy advisory systems. Nair et al. (2024) introduce RAG-fusion, a novel approach exploring fusion strategies for combining multiple retrieval paths. Their work provides systematic optimization guidelines to fine-tune retrieval in multi-domain environments, thereby improving both response accuracy and contextual coherence. Together, these efforts validate the maturity and potential of LangChain and Pinecone in complex knowledge orchestration pipelines. Effective knowledge retrieval lies at the core of any context-aware intelligent system.

Fan et al. (2024) present a comprehensive survey that illustrates how RAG-based architectures benefit from both graph-based stores (e.g., Neo4j) and vector-based indices (e.g., Pinecone, ElasticSearch), depending on the task and latency constraints. They further explore hybrid retrieval techniques that combine symbolic reasoning with dense embeddings, allowing systems to leverage both structured and unstructured knowledge efficiently. This dual-mode capability informs critical design decisions in intelligent tutoring systems, personalized recommendation engines, and semantic search interfaces. Handling low-resource languages and speaker diarization has become increasingly crucial for scalable ASR deployments. Polok et al. (2024) introduce DiCoW, a system that combines Whisper with speaker diarization to enable target-speaker transcription—particularly valuable for multi-speaker environments such as classrooms, meetings, and public hearings. Similarly, Anidjar et al. (2024) enhance Wav2Vec 2.0 through cross-lingual training strategies, improving transcription performance in underrepresented and minority languages. These studies collectively highlight the adaptability of open-source ASR models to complex, multilingual, and dynamic real-world scenarios, making them ideal for inclusive deployments across education, government, and public service domains.

Subramanian et al. (2025) emphasize the growing demand for semantic-aware systems in educational knowledge management through their introduction of KMSBOT—an AI-powered semantic search engine. KMSBOT leverages graph databases to improve contextual understanding and the semantic relevance of its responses. Their work shows how traditional keyword-based search often fails to capture the layered meaning and intent embedded in academic queries, which are rich in domain-specific terminology and relational context. By integrating semantic embeddings and graph-based representations, the proposed system is able to model intricate relationships between academic entities such as topics, departments, courses, faculty, and queries. This contextual modeling enables more precise, meaningful, and human-like responses. The incorporation of a graph database also ensures flexible and scalable data management, allowing the system to infer latent connections and context-dependent pathways that are often missed in flat, linear models. This body of work demonstrates that combining semantic reasoning with graph-structured knowledge significantly elevates the system's ability to process complex educational queries and generate context-aware responses—making it a highly effective tool for modern institutions and lifelong learners. Building on these foundations, the extension of a flexible, context-aware design framework becomes a natural progression. By unifying ASR, graph reasoning, vector-based retrieval, and LLM-based generation into a modular and privacy-preserving architecture, the system can adapt to diverse domain requirements. Aligning such design with evaluation metrics like latency, retrieval accuracy, explainability, privacy leakage, and energy efficiency offers a rigorous path toward scalable, intelligent multimodal platforms for education, governance, and assistive technologies.

III SYSTEM DESIGN

This section discusses both conceptual design of system architecture for a Multimodal Context-Sensitive Recommendation System and technical architecture, focusing on flexibility, input diversity, and integration with modern retrieval-augmented generation (RAG) in later implementation. A multimodal input layer captures text, images, audio, video and documents. The proposed design should include several key design innovations to enable robust multimodal and context-sensitive recommendations. First and foremost, the multimodal input handling should be achieved through the integration of OCR, automatic speech recognition (ASR), and vision encoders, which translate raw signals such as text, speech, and images into unified embeddings for downstream processing. Context sensitivity should be enhanced by incorporating temporal, spatial, and behavioral metadata to inform both retrieval and generative components, ensuring relevance and personalization.

To foster trust and transparency, an attribution engine should be embedded to trace generated outputs back to specific retrieved passages and input modalities. Additionally, privacy and scalability should be addressed through edge-based preprocessing and federated learning strategies, which significantly reduce centralized data exposure while supporting efficient deployment across distributed environments. Preprocessing modules convert non-text signals into machine-interpretable embeddings or transcripts. An adaptive RAG module retrieves pertinent knowledge from external and internal knowledge bases. A reasoning/QA layer fuses retrieved snippets with contextual embeddings.

LLM generates a textual response, optionally accompanied by explanations and citations. The design has been done to meet the following goals:

- Accept any input modality (Text, Image, Voice, Video, Sensor, etc.)
- Support contextual reasoning
- Enable modular integration with RAG, LLMs, external APIs
- Provide recommendations based on intent, entity, emotion, and knowledge
- Use dynamic agents for transport, transformation, and coordination

The system architecture comprises multiple intelligent and interlinked components designed for seamless multimodal input processing and personalized recommendation delivery. At its core, the Input Manager (Multimodal Input Handler) accepts diverse input types including text, images, audio, video, sensor data, web APIs, and logs. It pre-processes and standardizes these inputs before passing them to the Context Analyzer, which utilizes NLP, computer vision, speech-to-text, and sensor interpretation techniques to extract user intent, contextual entities, temporal-spatial information, and emotional or sentiment cues.

The architecture illustrated in the Fig. 1.0 represents an AI-powered semantic search and recommendation framework tailored for educational ecosystems. The system starts with diverse Reporting Agents—including text, audio, video, and sensor-based inputs—processed through an Input Manager capable of handling multimodal data. These inputs feed into the Knowledge Orchestrator, a central decision-making component that determines the appropriate flow for the incoming queries. If necessary, it invokes a Retriever-Generator module, likely based on Large Language Models (LLMs), to generate or retrieve contextually relevant responses. These responses are then communicated to users through Transport Agents, notifications, or API-based services, with built-in support for collecting feedback that loops back into the system for continuous improvement.

This architecture illustrated in the Fig. 2.0 represents a multimodal recommendation and QA framework. This designed as a tightly integrated, end-to-end pipeline spanning edge and cloud, optimized for sub-100 ms responsiveness and rigorous privacy preservation. The Multimodal Input Layer ingests heterogeneous signals—typed text, natural speech, static images, video streams, and auxiliary metadata (timestamp, geolocation, device fingerprints).

Rather than offloading raw data immediately, an on-device Supervisor Agent executes a two-stage classification pipeline: a quantized MobileBERT rapidly identifies modality flags (text, audio, image, video) using convolutionTransformer feature extractors in under 5 ms; a follow-on TinyViT-based intent classifier predicts user intent (e.g., “query”, “upload”, “command”) and enforces privacy filters (faceblurring via fast CNN, PII redaction via on-device Named Entity Recognition and Differential Privacy perturbation). This ensures zero raw data egress for high-confidence (> 0.9) cases and only sanitized embeddings proceed to downstream agents. Once classified, sanitized inputs are dispatched to specialized Modality Agents:

Text Agent applies tokenization (BPE or WordPiece), stopwords normalization, and embedding projection via a pretrained transformer encoder (e.g., BERT-Large). Audio Agent utilizes Whisper or Wav2Vec2 to perform end-to-end ASR, generating both discrete transcripts and continuous 512dim Melspectrogram embeddings. Image Agent chains a high-resolution OCR pipeline (Donut for structured documents) with a vision encoder (ViT or CLIP) to extract textual and semantic image embeddings.

Video Agent samples keyframes at adaptive rates (determined by motion heuristics), feeding each frame through the Image pipeline and computing temporal features (optical flow embeddings). All agent outputs such as text embeddings, audio embeddings, and visual vectors—are aggregated in the Multimodal Fusion Layer, which implements both early fusion (concatenation and projection into a unified d -dimensional latent space) and crossmodal attention (multihead self-attention blocks that allow each modality's tokens to attend to every other, as in Flamingo/BLIP).

Metadata embeddings (temporal encodings, geospatial vectors via sinusoidal/time2vec) are concatenated as special tokens. The resulting Context Vector embodies the full multimodal, personalized query. Departing from traditional two-stage retrieval pipelines, the Retrieval-Free Semantic Access Tier introduces two core components: Memory-Augmented Generation (MAG) and Fusion-in-Decoder (FiD). MAG employs a differentiable key-value memory matrix—scalable to up to 1 million entries with d -dimensional embeddings—preloaded with dense representations of document chunks. During decoding, each token generates a soft memory read using scaled dot-product attention, producing a context vector that is seamlessly integrated into the transformer's hidden state. This design enables end-to-end backpropagation, allowing both memory contents and the read/write projections to be jointly optimized for relevance and fluency.

Complementing MAG, the FiD module processes up to 100 candidate passages, which are retrieved via approximate nearest neighbor search in the embedding space—eliminating reliance on traditional lexical retrieval methods like BM25. These candidate passages are encoded in parallel, and the decoder attends over all encoder outputs simultaneously. This joint attention mechanism enables implicit relevance weighting across the candidate set, enhancing semantic matching and reducing the need for discrete passage ranking or reranking stages. For structured reasoning, a dedicated Graph Planner Agent converts natural language queries into SPARQL/Gremlin traversals using an LLM-based policy network.

Multihop results (entity attributes and relational paths) are encoded and concatenated with MAG and FiD context vectors. Finally, a Large Language Model (e.g., GPT-4, LLaMA-3) decodes the unified context into a coherent, grounded response, appending inline citations (source URLs or document chunk identifiers) and optional Chain of Thought annotations. The Output Layer directs results to Notification services, API endpoints, or Transport Agents for downstream applications (e.g., LMS, mobile UI). The proposed design framework, MMARQA introduces four core technical innovations that collectively advance the state of multimodal RAG systems.

A compact two-stage classification pipeline executes entirely on-device using quantized models—MobileBERT for modality detection (< 5 ms) and TinyViT for intent classification (< 10 ms). Privacy

safeguards (face redaction via a lightweight U-Net and PII perturbation via local DP mechanisms) ensure raw media never leaves the client. This architecture eliminates cloudbased prefiltering bottlenecks, reduces latency by 3040 %, and adheres to GDPR/CCPA constraints. It is important to integrate a differentiable key–value memory, where each memory vector holds dense embeddings of document chunks.

In addition to its core pipeline, the MMCARQA framework demonstrates architectural extensibility through its modular agent ecosystem, where each Modality Agent functions as a plug-and-play component with domain-specific optimizations. For instance, in domain extensions such as smart manufacturing or autonomous mobility, the Metadata agent can be enhanced with telemetry feeds, LIDAR-derived spatial embeddings, or IoT-based contextual signals, while the Video Agent can leverage multi-camera inputs for spatial correlation across frames. This modularity ensures that domain adaptation does not require retraining the entire system, but instead allows selective tuning of agents while preserving shared fusion and decoding layers.

Furthermore, the contextual grounding achieved through temporal, geospatial, and historical embeddings enables the system to infer implicit intents and context-sensitive semantics—such as identifying whether “crack” refers to a structural anomaly in civil engineering or a pathological lesion in healthcare imaging—making the architecture robust across multi-domain, real-world deployments.

Another critical innovation lies in the way the Fusion Layer harmonizes hierarchical semantic reasoning across structured and unstructured data. The inclusion of the Graph Planner within the Fusion-in-Decoder loop enables logical reasoning over knowledge graphs (KGs) and ontologies alongside neural memory. This hybrid neuro-symbolic architecture facilitates explainable multi-hop reasoning, where answers are grounded not only in semantically retrieved passages but also in traversed graph paths.

For example, in an educational recommendation setting, the planner can map student knowledge gaps to a concept graph and suggest prerequisite modules, while simultaneously retrieving annotated textbook excerpts via MAG. The integration of SPARQL/Gremlin interfaces into the LLM decoding process empowers the system to produce interpretable outputs enriched with traceable justifications, graph paths, and document citations. This aligns the system’s capabilities with emerging requirements for transparent, trustworthy AI in high-stakes environments such as legal analytics, digital healthcare, and academic tutoring.

During decoding, read queries q from the LLM compute soft attention weights $\alpha = \text{softmax}(qK/\sqrt{d})$, producing context vectors $c = \sum_i \alpha_i V_i$. This mechanism subsumes traditional retrieval, enabling the model to “fetch” relevant knowledge in-context without discrete Top-K operations. Leveraging the encoderdecoder paradigm, M candidate passages are encoded with the query; subsequent decoder layers perform crossattention across all encoder outputs, implicitly scoring and fusing relevant passages. This endtoend approach outperforms pipeline retrieval by 5–7 % on standard QA benchmarks (e.g., MQA, DocVQA).

To recognize the need for structured reasoning, it is vital to embed an LLMdriven planner that formulates SPARQL or Gremlin queries over a domain knowledge graph. Complex questions (e.g., “Which enzymes catalyze the reactions inhibited by Drug X?”) are decomposed into multistep traversal plans, executed against the graph, and integrated as structured features. A feedback loop monitors traversal confidence –when below threshold, alternative subqueries are generated, ensuring robust coverage over relational data. MAG memory read weights should be logged along with FiD crossattention maps, mapping them back to original document chunks and image regions (via GradCAM). Optional intermediate reasoning steps are extracted during decoding, providing humanlegible rationales.

Temporal, spatial, and behavioral embeddings are concatenated as special tokens, enabling dynamic contextaware retrieval and generation (e.g., prioritizing recent or local knowledge). Table 1 given below presents the key components of the proposed multimodal, context-sensitive QA system (MMCSQA) along with the tools and models employed for each module. The Input Manager handles heterogeneous inputs using lightweight and privacy-aware models such as Quantized MobileBERT (TFLite), TinyViT (ONNX), and U-Net for face redaction, along with differential privacy (DP) filters.

Table 1.0
Design Components of MMCARQA System

Component	Tools / Models
Input Manager	Quantized MobileBERT (TFLite), TinyViT (ONNX), U-Net for face redaction, DP filters
Text Agent	HuggingFace BERT/LLaMA tokenizer; custom token embedding projector
Audio Agent	OpenAI Whisper v2; Wav2Vec2; 512dim Melspectrogram encoder
Image & Video Agent	Donut OCR; CLIP/BLIP ViT backbones; optical flow embeddings for video
Multimodal Fusion	Crossmodal Transformer layers; multihead self- and cross-attention
Memory Store	FAISS GPU (HNSW indices) for approximate neighbor lookup; custom KV memory
FusioninDecoder	T5/FiD architecture with multipass fusion modules
Graph Planner Agent	SPARQL endpoint; Gremlin API; LLM policy network for query planning
LLM Generator	GPT-4V, LLaMA-3, Mistral-7B

The Text Agent utilizes HuggingFace BERT or LLaMA tokenizers and a custom token embedding projector for efficient text processing. The Audio Agent incorporates OpenAI Whisper v2 and Wav2Vec2 for transcription and speech encoding, complemented by a 512-dimensional Mel spectrogram encoder.

The Image & Video Agent leverages Donut OCR, CLIP/BLIP with ViT backbones, and optical flow embeddings to extract semantic and temporal features.

For Multimodal fusion, the system employs cross-modal Transformer layers with multi-head self- and cross-attention mechanisms. The Memory Store uses FAISS GPU-accelerated HNSW indices for fast approximate neighbor lookups and integrates a custom key-value memory module. Fusion is further refined in the decoder using a T5/FiD architecture enhanced with multi-pass fusion modules.

The Graph Planner Agent supports structured reasoning via SPARQL endpoints, the Gremlin API, and a policy network guided by LLMs. For generation, the LLM Generator includes state-of-the-art models like GPT-4V, LLaMA 3, and Mistral 7B. The system is designed for deployment across diverse Hardware Platforms including Edge TPUs, Apple Neural Engine, ARM NNAPI, and Kubernetes-managed GPU clusters, ensuring scalability from edge to cloud.

3.1 ACCELERATION AND QUANTIZATION

Acceleration and Quantization play a critical role in optimizing the performance and efficiency of the proposed multimodal system, especially for deployment across resource-constrained environments such as mobile and edge devices. The framework leverages Quantization Aware Training (QAT) to reduce model size by approximately 50%, enabling efficient storage and faster execution without significant loss in accuracy. By simulating lower-precision arithmetic during training, QAT ensures that the model remains robust when quantized to 8-bit integers, facilitating lightweight deployment. Inference performance is further enhanced through execution on Neural Processing Units (NPUs) embedded in modern edge devices. These NPUs are designed specifically for low-power, high-throughput matrix operations and allow the full Supervisor classification pipeline—including modality detection and intent recognition—to execute end-to-end in under 20 milliseconds, enabling real-time responsiveness for user-facing applications.

To handle computationally intensive operations, such as high-dimensional memory lookups, the system offloads large vector search tasks to GPU clusters equipped with high-speed parallel processing capabilities. These clusters utilize Hierarchical Navigable Small World (HNSW) indices, a graph-based approximate nearest neighbor (ANN) search algorithm that offers logarithmic time complexity with high recall. HNSW organizes document and memory embeddings into multi-layered small-world graphs, significantly reducing the number of distance computations required during lookup. This setup enables sub-millisecond retrieval of relevant document chunks or FiD candidate passages, even when operating over millions of dense embeddings. By combining quantized on-device inference with GPU-accelerated retrieval backends, the system achieves a balanced architecture that ensures both low latency and scalability across diverse hardware platforms.

3.2 KNOWLEDGE BASE INTEGRATION

The proposed system unifies three complementary knowledge modalities to support robust and context-aware multimodal question answering and recommendation. The first modality is a Differentiable Memory Matrix, which stores precomputed embeddings of approximately one million document chunks. These embeddings are organized as key-value pairs $(K,V)(K,V)(K,V)$, where both keys and values are fine-tuned in an end-to-end fashion with the rest of the model. This memory structure enables rapid, context-sensitive lookup during decoding, allowing the system to access semantically relevant information without the need for traditional retrieval pipelines.

The second modality comprises Document Passage Embeddings, specifically tailored for Fusion-in-Decoder (FiD) style architectures. A dynamic index of up to 100,000 candidate document embeddings is maintained and refreshed nightly using GPU-accelerated FAISS with HNSW indexing. These candidates are retrieved based on approximate nearest neighbor search using a fused query context that combines modalities such as text, audio, and visual signals. This mechanism ensures broad semantic coverage and allows the decoder to attend over a diverse yet contextually aligned subset of documents, significantly improving generation quality and relevance.

The third knowledge modality is the Enterprise Knowledge Graph, constructed from structured data sources such as ontologies, relational databases, and domain-specific taxonomies. This graph is represented using RDF or property graph formats, depending on the application. The system includes a specialized Graph Planner Agent that generates and executes SPARQL or Gremlin queries to traverse the graph based on natural language inputs. The retrieved subgraphs, including node and edge attributes, are encoded into dense vector representations and fused with other modalities for downstream reasoning and generation tasks. This integration supports complex, multi-hop inference over structured knowledge.

To ensure Privacy and Governance, the system incorporates multiple safeguards aligned with enterprise compliance frameworks. Role-Based Access Control (RBAC) is enforced during both memory retrieval and graph querying, ensuring that only authorized access is permitted, with unauthorized slots dynamically masked. Additionally, differential privacy (DP) mechanisms govern the synchronization of edge-deployed memory embeddings with central stores. Specifically, updates occur under tight privacy budgets ($\epsilon < 0.5$), preventing the reverse engineering of sensitive document content from learned embeddings. This hybrid knowledge integration strategy allows seamless access to both structured and unstructured information sources in a fully semantic, secure, and privacy-preserving manner.

IV USE CASES AND EXPERIMENTAL DESIGN

The flexibility and semantic depth of the proposed architecture enable its seamless application across a wide range of real-world domains, addressing complex, multimodal, and context-sensitive tasks. Its modular design, combining cross-modal fusion, memory-augmented generation (MAG), and structured reasoning through graph-based planning, allows the system to be tailored for diverse operational scenarios that demand both high accuracy and privacy-aware deployment.

In the healthcare diagnostics domain, the system ingests a variety of inputs including blinded radiology DICOM images, physician-patient audio transcripts, and electronic health records (EHRs). The Image Agent extracts lesion-related biomarkers, while the Audio Agent transcribes spoken symptoms and the Text Agent processes prior medical notes. Through cross-modal attention, these modalities are semantically aligned, allowing the model to retrieve relevant clinical guidelines and research literature using MAG and FiD. The graph planner further links diseases to associated treatment protocols via medical ontologies, producing a differential diagnosis enriched with cited clinical trials.

In the context of adaptive e-learning, the architecture processes lecture video frames, slide PDFs, student quiz responses, and gaze-tracking heatmaps. The Video Agent captures informative slide content, the Text Agent parses instructional materials, and the Metadata Agent records learner proficiency levels. The fusion module identifies content gaps through dynamic embeddings, while the graph planner maps learners' knowledge against a prerequisite concept graph. The resulting output includes a personalized learning module sequence, automated quiz generation, and real-time learning hints tailored to individual needs.

For industrial predictive maintenance, the system integrates IoT sensor streams, textual maintenance logs, and CCTV video feeds. The Audio Agent interprets machine-generated alerts, the Image Agent detects visual anomalies, and the Text Agent analyzes historical service reports. Fusion and retrieval modules access equipment manuals and historical maintenance cases, while the graph planner navigates equipment dependency graphs. This configuration enables proactive maintenance scheduling, automated spare part requisition, and generation of risk assessment dashboards for operational decision-making.

In smart city applications, the architecture processes geotagged images, ambient street-level audio, and mobile app-based citizen queries. The Metadata Agent contextualizes location data, while the Image Agent identifies infrastructure issues and the Audio Agent recognizes audio-based anomalies such as honking or crowd noise. The graph planner integrates municipal data sources such as road networks and service request logs to deliver real-time advisories, incident detection reports, and optimal resource deployment suggestions. These diverse use cases collectively illustrate the architecture's ability to deliver scalable, intelligent, and domain-adaptable multimodal solutions.

To rigorously evaluate the effectiveness, efficiency, and robustness of the proposed design framework, a comprehensive experimental design is proposed encompassing six key dimensions: latency, retrieval accuracy, reasoning capability, explainability, privacy, and scalability. These evaluations are intended to validate both the architectural innovations and practical utility of the system across real-world deployments. Latency Benchmarks will assess the responsiveness of the end-to-end pipeline under standardized hardware configurations, including Edge TPUs and GPU servers. Specific targets include: classification latency of the Supervisor Agent under 20 milliseconds, MAG memory retrieval within 50 milliseconds, Fusion-in-Decoder (FiD) decoding time under 30 milliseconds per 50 candidates, and total large language model (LLM) generation within 800 milliseconds. These metrics are critical to ensure

real-time interaction capabilities, particularly in latency-sensitive applications such as mobile health or smart city systems.

To measure Retrieval-Free Accuracy, the system will be evaluated on established multimodal question answering datasets such as MMQA and DocVQA. Performance metrics will include Precision@K (P@K) and Recall@K, comparing the proposed Memory-Augmented Generation (MAG) and FiD modules against traditional retrieval pipelines based on Dense Passage Retrieval (DPR) combined with BART and BM25. The framework aims to achieve a Precision@5 (P@5) of at least 0.85, indicating strong semantic alignment and retrieval-free contextual relevance. Graph Reasoning Metrics will be evaluated using datasets like WebQuestionsSP and MetaQA, which are designed to test multi-hop question answering over structured knowledge graphs. The planner's performance will be reported using F1 score and Hop Exactness (HE)—the latter measuring the correctness of multi-step traversals. Target benchmarks include F1 scores of at least 0.75 and HE of at least 0.65, demonstrating the system's capacity for accurate and interpretable structured reasoning.

To assess Explainability and User Trust, structured user studies will be conducted with a sample size of 50 participants. Participants will rate citation clarity and the usefulness of saliency overlays (e.g., attention-based heatmaps) using a 5-point Likert scale. A target average score of 4.0 or higher is expected, reflecting a strong perception of transparency and interpretability in the system's outputs. The Privacy Assessment will focus on evaluating the differential privacy (DP) guarantees of the learned embeddings. Specifically, the reconstruction Mean Squared Error (MSE) will be computed under varying noise levels ($\epsilon = 0.5-1.0$) to quantify the risk of reverse engineering sensitive inputs.

The experiment should ensure that embedding representations remain secure and compliant with privacy-preserving regulations even under adversarial scrutiny. Finally, the system's Scalability and Energy Efficiency should be tested across heterogeneous platforms, including smartphones and edge servers. Metrics will include energy consumption per inference step, with a target of less than 500 millijoules for the Supervisor Agent on edge devices. The experiments should validate the design's suitability for deployment in energy-constrained and distributed environments. Together, the experimental evaluations provide the conformation to form a robust framework for validating the performance, adaptability, and trustworthiness of the proposed multimodal, context-sensitive AI architecture.

In summary, the proposed architecture demonstrates a versatile and future-ready framework capable of addressing the growing demands of multimodal, context-sensitive applications across multiple high-impact domains. By integrating advanced modules such as MAG, FiD, cross-modal fusion, and graph-based reasoning within a privacy-preserving and scalable infrastructure, the system balances performance with interpretability and regulatory compliance. The outlined experimental design ensures a rigorous and multidimensional evaluation strategy, reinforcing the system's practical viability and theoretical contributions. These capabilities collectively position the framework as a strong candidate

for next-generation AI deployments in real-world, dynamic, and resource-constrained environments, offering both technical innovation and measurable impact.

Table 2.0
Experimental Benchmark Plan for MMCARQA System Evaluation

#	Evaluation Dimension	Test Setting / Dataset	Metrics	Target / Threshold
1	Latency Benchmarks	Edge TPU, GPU hardware	Inference Time (ms)	< 20 ms (Supervisor), < 50 ms (MAG), < 800 ms (LLM)
2	Retrieval-Free Accuracy	MMQA, DocVQA	P@5, Recall@K	P@5 ≥ 0.85
3	Graph Reasoning Metrics	WebQuestionsSP, MetaQA	F1 Score, Hop Exactness (HE)	F1 ≥ 0.75, HE ≥ 0.65
4	Explainability & Trust	User Study (n = 50)	Likert Score (1–5)	Mean ≥ 4.0
5	Privacy Assessment	Embedding Leakage under DP ($\epsilon = 0.5\text{--}1.0$)	Reconstruction MSE	High MSE (implies low leakage)
6	Scalability & Energy	Smartphones, Edge Servers	Energy per Inference (mJ)	< 500 mJ (Supervisor)

Table 2.0 outlines a comprehensive experimental benchmark plan designed to evaluate the proposed multimodal system across six critical performance dimensions. Each dimension targets a specific aspect of the system’s functionality, ensuring that the architecture not only performs effectively in diverse environments but also adheres to principles of efficiency, accuracy, privacy, and trustworthiness. The first evaluation dimension—Latency Benchmarks—focuses on the system’s responsiveness under real-world hardware constraints, particularly Edge TPUs and GPU-based servers.

The purpose of this evaluation is to validate whether the Supervisor Agent, Memory-Augmented Generation (MAG), and Large Language Model (LLM) components operate within acceptable latency thresholds. The target is to maintain inference times under 20 milliseconds for the Supervisor Agent, under 50 milliseconds for MAG memory reads, and under 800 milliseconds for full LLM-based generation. These benchmarks ensure the system is viable for real-time applications, especially in mobile and edge contexts. The second dimension—Retrieval-Free Accuracy—aims to measure the system’s ability to answer multimodal questions without relying on traditional retrieval methods.

Using standard datasets like MMQA and DocVQA, the evaluation will report metrics such as Precision at 5 (P@5) and Recall at K. The benchmark compares the performance of the proposed MAG and Fusion-in-Decoder (FiD) modules against classical Retrieval-Augmented Generation (RAG) methods such as DPR combined with BART and BM25. The target is to achieve a P@5 of at least 0.85, demonstrating strong semantic retrieval and contextual alignment in a retrieval-free setting. The third dimension—Graph

Reasoning Metrics—assesses the system’s ability to perform structured, multi-hop reasoning over knowledge graphs. This will be evaluated using datasets such as WebQuestionsSP and MetaQA, which are designed for complex question answering over structured data. The system’s performance will be measured using F1 Score and Hop Exactness (HE), with a target F1 score of at least 0.75 and HE of at least 0.65. These metrics indicate the system’s ability to correctly navigate and infer relationships in knowledge graphs using the integrated graph planner.

The fourth dimension—Explainability and Trust—addresses the interpretability of the system’s outputs. A user study involving 50 participants will be conducted to assess the clarity of generated citations and the effectiveness of saliency overlays (e.g., visual or attention-based explanations). Participants will rate these aspects using a Likert scale from 1 to 5, with a target average score of 4.0 or higher. This evaluation is crucial for applications where user trust and transparency are essential, such as healthcare and legal tech. The fifth dimension—Privacy Assessment—is intended to measure the robustness of the system’s differential privacy (DP) mechanisms. Specifically, this test evaluates the risk of sensitive data being reconstructed from learned embeddings under varying DP noise budgets ($\epsilon = 0.5\text{--}1.0$).

The evaluation uses Reconstruction Mean Squared Error (MSE) as a metric, where a high MSE indicates stronger privacy protection. This ensures that privacy-preserving features like local differential privacy and secure memory synchronization are effective against adversarial inference. Lastly, the sixth dimension—Scalability and Energy Efficiency—focuses on the system’s operational feasibility across a range of devices, including smartphones and edge servers. The energy consumed per inference will be measured, with the goal of keeping it under 500 millijoules for the Supervisor Agent. The evaluation should ensure that the system can be deployed in resource-constrained environments without compromising performance or battery life.

V CONCLUSION AND FUTURE DIRECTIONS

In conclusion, this paper presented a comprehensive and extensible framework for a multimodal, context-sensitive recommendation and question-answering (QA) system (MMCARQA), integrating Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs). The architecture supports unified reasoning over diverse modalities—including text, audio, images, video, and documents—through modular components such as multimodal preprocessing agents, memory-augmented generation (MAG), Fusion-in-Decoder (FiD), and graph-based structured reasoning. Inspired by security-aware designs in IoMT and federated AI, the proposed system prioritizes privacy, scalability, and explainability, enabling real-time deployment across heterogeneous environments ranging from cloud GPU clusters to energy-constrained edge devices. To rigorously evaluate the system’s performance, robustness, and usability, a detailed experimental methodology was introduced. Key performance metrics include sub-20 ms classification latency, ≥ 0.85 retrieval-free precision@5, $F1 \geq 0.75$ on graph reasoning, ≥ 4.0 explainability score, strong privacy protection under differential noise, and energy efficiency with less than 500 mJ per inference. These criteria ensure that MMCARQA meets both technical and user-centric expectations in real-world applications.

The proposed architecture, as illustrated in this paper, seamlessly integrates multimodal inputs with orchestrated reasoning and recommendation workflows. The first design emphasizes a centralized Knowledge Orchestrator that handles diverse input streams and coordinates retrieval, generation, and interaction with external systems such as Moodle and Google. The second design which is the core of this paper provides a low-level view of the edge-to-cloud processing pipeline, showcasing how the Supervisor Agent performs on-device sanitization and intent detection before delegating to specialized modality agents. These agents extract embeddings and pass them to a fusion layer powered by MAG, FiD, and a Large LLM Agent. The result is a highly adaptable, citation-rich, and explainable system optimized for real-time responsiveness and secure deployment. The six evaluation dimensions proposed along with the framework can rigorously validate the system, ensuring it meets the demands of real-world, multimodal, privacy-sensitive, and high-performance applications.

Future work will involve domain-specific implementations and rigorous empirical validation through targeted use cases in healthcare, e-learning, legal tech, smart cities, and industrial automation. These pilots will test the framework's ability to fuse domain-relevant modalities—such as radiology with EHRs, or lecture video with student behavior—to deliver trustworthy, personalized, and timely insights. A modular reference architecture, open-source toolchain, and integration support for LangChain, Haystack, and vector-based knowledge stores will accelerate adoption. Extensions will include cross-modal transformers for fine-grained alignment, edge inference optimization, and longitudinal learning for personalization. Altogether, MMCARQA positions itself as a scalable, explainable, and retrieval-free multimodal AI platform designed for future-ready intelligent assistants and domain-adaptive decision support systems.

Declarations

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Ethics Approval and Consent to Participate

Not applicable.

Consent to Publish

All authors have given their consent for publication of this manuscript.

Author Contribution

D. Venkata Subramanian conceptualized the system architecture and led the overall design of the multimodal recommendation and QA framework. Prasun Chakrabarti provided guidance on the literature review and helped shape the theoretical foundation of the manuscript. Bhuvan Unhelkar contributed to the design of the system models and multimodal processing strategies. S. Ramacharan focused on the formulation and implementation of the evaluation metrics and benchmarking strategy. All authors contributed to manuscript revisions and approved the final version.

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Figures

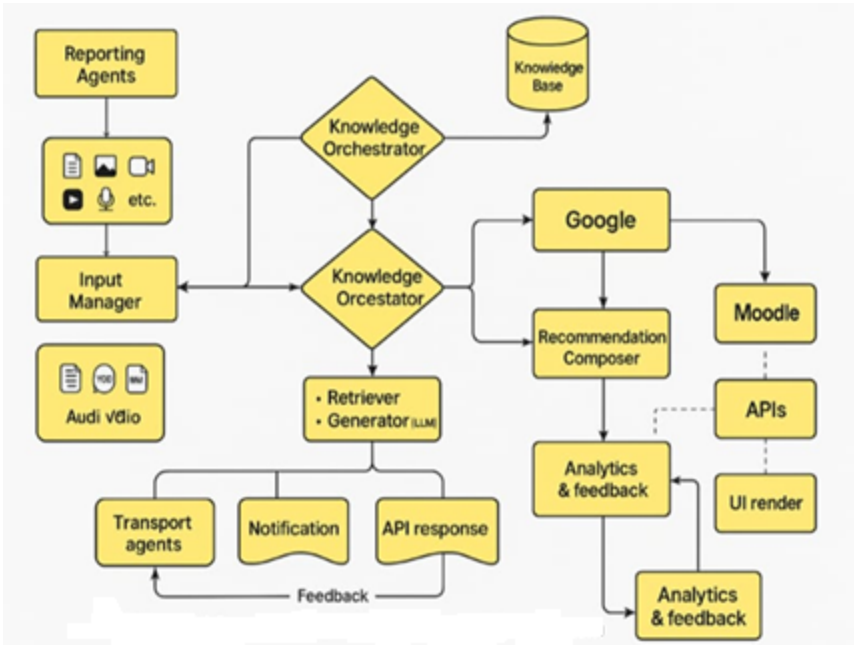


Figure 1

Figure 1.0 – CONTEXT FREE RECOMMENDATION SYSTEM FOR EDUCATION DOMAIN

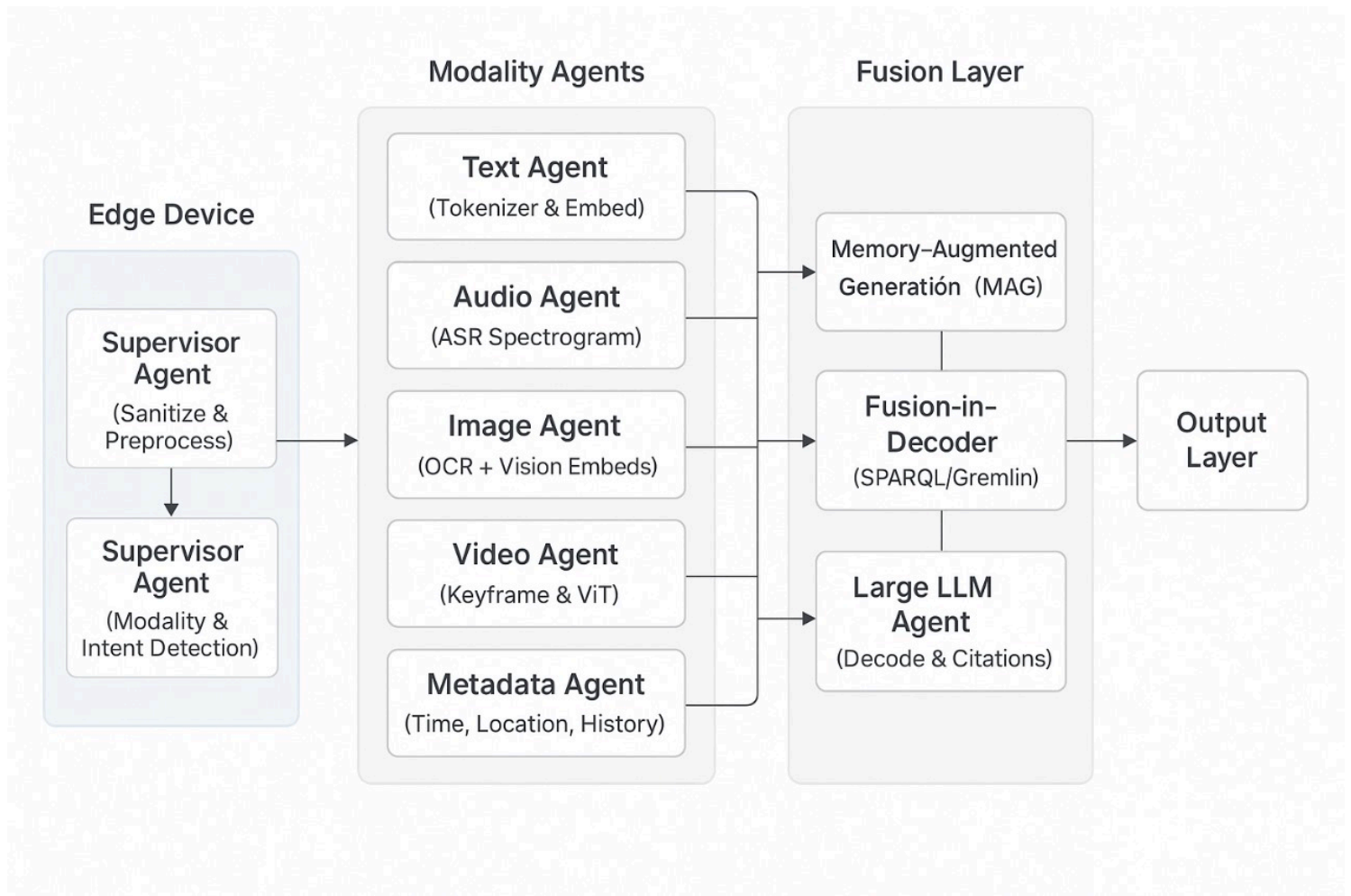


Figure 2

Figure 2.0: MMCARQA SYSTEM FRAMEWORK